

TRANSFORMING HIGHER EDUCATION IN SUB-SAHARAN AFRICA THROUGH ARTIFICIAL INTELLIGENCE: A SCOPING REVIEW

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Abstract

This scoping review investigates the role of Artificial Intelligence (AI) in higher education institutions in sub-Saharan Africa, emphasizing the strategic objectives, advantages, obstacles, and ethical implications of its implementation. This study synthesises insights from ten pivotal articles derived from an extensive bibliometric analysis of 149 publications available in the Scopus database, covering the period from 2016 to 2024. This review underscores the potential of AI to substantially improve educational quality by facilitating personalized learning experiences and enhancing operational efficiency. The incorporation of AI is aligned with various Sustainable Development Goals (SDGs), particularly Goal 4: Quality Education, as it improves access to quality educational resources and encourages innovative teaching methodologies. Nevertheless, significant challenges such as inadequate infrastructure, resistance from faculty, and lack of training programs impede effective integration. Ethical issues concerning data privacy and algorithmic bias were also examined, highlighting the necessity for comprehensive guidelines to promote responsible AI utilisation in educational settings. The findings are intended to guide policymakers and educators in adopting the best practices for AI integration in higher education, considering the distinct socio-economic context of sub-Saharan Africa.

Keywords: Artificial intelligence in education, Sustainable development goals, Higher education integration.

1 INTRODUCTION

The role of artificial Intelligence (AI) in higher education has attracted considerable global interest, especially in sub-Saharan Africa where educational obstacles are particularly severe [1]. AI applications can potentially improve academic outcomes by facilitating personalized learning pathways and enhancing the operational efficiency of educational institutions [2]. Nonetheless, the effective deployment of AI is frequently obstructed by inadequate infrastructure and the reluctance of educators to doubt its efficacy [3; 4]. Ethical issues are paramount in AI integration, particularly regarding data privacy and the risk of algorithmic bias, which underscores the need for comprehensive guidelines to ensure responsible implementation [5; 6]. This review examines these aspects to guide future initiatives that integrate AI into sub-Saharan Africa's higher education framework.

The objectives of this study are as follows:

- 1 Identify the strategic goals and benefits of integrating AI into higher education institutions in Sub-Saharan Africa. This includes exploring how AI can enhance educational outcomes, promote personalised learning, and drive innovation in academic practice
- 2 Investigate the challenges and barriers faced by higher education institutions in sub-Saharan Africa in adopting and implementing AI. This objective aims to uncover the socio-economic, infrastructural, and technological challenges to be addressed for successful AI integration.
- 3 Examine the ethical considerations in using AI in higher education. This includes addressing concerns regarding data privacy, equity in access to AI-driven educational tools, and the broader impact of AI on educational landscapes.

2 METHODOLOGY

Search Strategy

(TITLE-ABS-KEY ("Artificial Intelligence" OR "AI") AND TITLE-ABS-KEY ("Higher Education" OR "University" OR "College" OR "Institution") AND TITLE-ABS-KEY ("STEM Education" OR "Science Education" OR "Technology Education" OR "Engineering Education" OR "Mathematics Education"

) AND TITLE-ABS-KEY ("Integration" OR "Implementation" OR "Adoption") AND TITLE-ABS-KEY ("Benefits" OR "Challenges" OR "Ethical Considerations" OR "Strategic Goals"))

Dataset pre-processing and refinements

This study implemented a systematic pre-processing protocol to ensure the relevance and manageability of the dataset. The refinement process included the following steps:

- Removal of duplicates
- Screening of titles, keywords, and abstracts to ensure direct relevance to the study topic
- Application of filtration criteria
- Temporal scope: 2016-2024
- Document types: peer-reviewed articles, book chapters, reviews, and conference proceedings
- Language: English-language publications only

This refinement process resulted in a final dataset of 149 articles that were pertinent to our research scope.

The results of our data collection and refinement process are visually represented in Fig. 1, which follows the PRISMA guidelines for systematic reviews. The PRISMA flow diagram presents the results of the following studies.

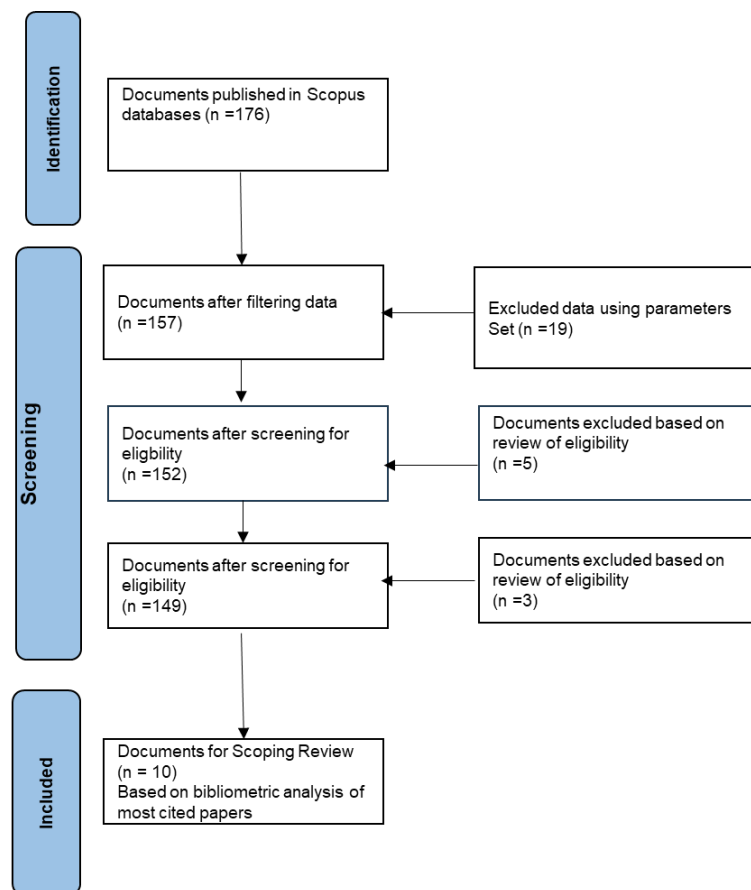


Fig. 1 Processing of document eligibility based on PRISMA guidelines.

3 RESULTS

In this section, we outline the findings of our comprehensive review and provide a context for the emerging academic discourse on revolutionising higher education in sub-Saharan Africa through the application of artificial intelligence and studies on pedagogical outcomes from 2016 to 2024.

3.1 Most Cited Publications: A bibliometric analysis of top publications shaping the discourse on transforming higher education in sub-Saharan Africa through artificial intelligence

A comprehensive understanding of the transformation of higher education in sub-Saharan Africa through artificial intelligence is essential. To this end, a rigorous bibliometric analysis of the most cited documents and their corresponding focus areas was conducted. From a pool of 149 extracted papers, 10 publications received over 20 citations each, as detailed in Table 1. A meticulous frequency count and thematic analysis of the keywords within these influential articles yielded the following key focus areas:

Table 1 Top Cited Publication (>200)

Document	Citations	Links
Sandu (2019)	106	0
Hernandez-de-menendez (2020)	79	0
Lin (2020)	74	0
Abulibdeh (2024)	71	0
Howell (2021)	69	0
Mclaughlin (2016)	65	0
Owoc (2021)	62	0
Kozak (2018)	28	0
Huang (2021)	23	0
Rajabi (2023)	21	0



Fig. 1 Network visualization of the ten key studies on AI integration in higher education

Table 2 summarises the ten significant research on incorporating artificial intelligence in higher education, emphasizing diverse methodologies and notable outcomes. An investigation conducted by [7] revealed that AI chatbots significantly improve student engagement and learning results by offering personalized assistance and feedback, which is especially advantageous in large classroom environments. [1] highlighted emerging trends in educational technology and advocated flexible teaching strategies to address future educational demands, underscoring the need for ongoing technology integration. [2] explored the contribution of AI towards achieving sustainable development goals, indicating a favorable relationship between AI adoption and enhanced educational practices. These studies underscore the urgent need for comprehensive policy recommendations that encompass infrastructure improvements, focus on educators' professional development, and create ethical frameworks to ensure responsible AI applications in educational contexts.

Table 2 Summary of Key Studies on AI Integration in Higher Education

Author(s)	Year	Title	Methodological Approach	Key Findings	Knowledge Gap	Policy Recommendation
Sandu N.; Gide E.	2019	Adoption of AI chatbots to enhance student learning experience in higher education in India	Mixed methodology	AI chatbots significantly improve student engagement and learning outcomes. They can provide personal support and feedback, enhancing the overall educational experience.	Limited research on long-term impacts	Encourage institutions to integrate AI technologies in curricula.
Hernandez-de-Menendez M.; Escobar Díaz C.; Morales-Menendez R.	2020	Technologies for the future of learning: state-of-the-art	Literature review	Identifies current trends in educational technology, emphasizing the importance of adaptability and innovation in teaching methods to meet future learning needs.	Lack of comprehensive frameworks	Develop policies that promote continuous technology integration in education.
Lin H.; Wan S.; Gan W.; Chen J.; Chao H.-C.	2022	Metaverse in Education: Vision, Opportunities, and Challenges	Qualitative	The metaverse presents unique opportunities for immersive learning experiences but poses challenges such as accessibility and technological readiness among educators and students.	Insufficient exploration of practical applications	Invest in training for educators on metaverse technologies.
Abulibdeh A.; Zaidan E.; Abulibdeh R.	2024	Navigating the confluence of artificial intelligence and education for sustainable development	Quantitative	Highlights the role of AI in achieving sustainable development goals through enhanced educational practices, suggesting a correlation between AI integration and improved educational outcomes.	Need for empirical data on effectiveness	Formulate strategies that align AI initiatives with sustainability goals.
Howell R.A.	2021	Engaging students in education for sustainable development: The benefits of active learning	Case study	Active learning strategies significantly enhance student engagement and retention, particularly in sustainability education, fostering critical thinking and problem-solving skills.	Limited focus on active learning methods	Promote active learning approaches across all educational levels.
McLaughlin J.E.; White P.J.; Khanova J.; Yuriev E.	2016	Flipped Classroom Implementation: A Case Report of Two Higher Education Institutions in the U.S.	Case study	Flipped classrooms lead to improved student performance and satisfaction compared to traditional teaching methods, highlighting the effectiveness of student-centered learning environments.	Need for broader implementation studies	Encourage universities to adopt flipped classroom models widely.
Owoc M.L.; Sawicka A.; Weichbroth P.	2021	Artificial Intelligence Technologies in Education: Benefits, Challenges and Strategies of Implementation	Mixed methodology	AI technologies can enhance educational outcomes but face challenges such as resistance from educators and lack of infrastructure; strategic implementation is crucial for success.	Underexplored educator perspectives	Develop training programs addressing concerns about AI integration in education.

Kozak S.; Ruzicky E.; Stefanovic J.; Schindler F.	2018	Research and education for industry 4.0: Present development	Literature review	Emphasizes the need for educational institutions to adapt curricula to meet the demands of Industry 4.0, focusing on interdisciplinary approaches and collaboration between academia and industry.	Insufficient alignment with industry needs	Foster partnerships between educational institutions and industries to co-develop curricula.
Huang C.; Yu K.	2021	Research on the innovation of college music teaching mode based on artificial intelligence	Qualitative	Innovative teaching modes using AI can significantly enhance music education by providing personalized feedback and adaptive learning pathways tailored to individual student needs.	Limited application in music education	Encourage exploration of AI tools specifically tailored for arts education.
Rajabi P.; Taghipour P.; Cukierman D.; Doleck T.	2023	Exploring ChatGPT's impact on post-secondary education: A qualitative study	Qualitative	ChatGPT has transformative potential in post-secondary education by facilitating personalized tutoring, enhancing student engagement, and providing instant feedback; however, challenges remain regarding its effective integration into existing curricula.	Need for quantitative assessments	Develop guidelines for effective ChatGPT integration into educational practices.

3.2 Strategic Goals, Challenges, and Ethical Considerations of AI In Higher Education

Table 3 presents the strategic goals, challenges, and ethical considerations for Artificial Intelligence in higher education.

3.2.1 Strategic Goals and Benefits of Integrating AI

Incorporating AI into educational institutions across sub-Saharan Africa aims to boost educational quality and streamline operations. A key advantage of this approach is the opportunity to tailor learning experiences. AI-powered tools, including chatbots, can bolster student support and improve engagement and academic performance. [7] showed that AI chatbots elevate students' educational experiences by providing prompt assistance and feedback, particularly valuable in crowded classrooms where individual attention is scarce. The integration of AI can also boost operational efficiency by automating administrative duties such as grading and scheduling. This increased efficiency allows educators to focus on teaching and mentoring students. [2] highlighted that AI tools can optimise processes and enhance resource management within institutions, ultimately leading to improved educational outcomes. AI supports developmental objectives by enhancing access to quality education and encouraging innovative teaching approaches. [1] asserted that the ongoing evolution of educational technology is essential for equipping students to face future challenges in the ever-changing job market.

3.2.2 Challenges and Barriers to Adoption

Despite its potential benefits, the successful integration of AI into higher education across sub-Saharan Africa faces several obstacles. The primary challenge is the inadequate infrastructure, which is prevalent in many institutions. The absence of stable internet connections and sufficient technological resources for effective AI implementation significantly hampers progress. [3] emphasised that these infrastructural limitations substantially impede the adoption of advanced technologies in educational contexts. Another significant hurdle is the reluctance of some educators to embrace AI. This resistance often stems from doubts about AI's effectiveness or concerns that it might render their positions obsolete. Such apprehension can be attributed to a lack of familiarity with AI and concerns about job security. To address this issue, [4] advocated targeted professional development and training initiatives aimed at

helping educators recognise the advantages and applications of AI in their teaching methodologies. Moreover, the scarcity of comprehensive AI technology training programs for faculties and students presents an additional barrier. The absence of adequate training can lead to difficulties in effectively utilising these tools, limiting their potential impact on educational outcomes [8].

3.2.3 Ethical Considerations

The incorporation of AI into higher education institutions creates ethical challenges. A primary concern is the protection of student data, as the collection and examination of this information by AI systems can compromise confidentiality and security. Educational establishments must implement strong safeguards to protect sensitive data from unauthorised use or access [5]. Another ethical consideration is the potential for AI systems to unintentionally reinforce existing prejudices if not properly designed and supervised; this is known as an algorithmic bias. This issue calls for ethical frameworks to ensure the equitable and transparent use of AI in educational settings [6]. There is also the risk that excessive reliance on technology might impair students' ability to think critically and solve problems independently if they become overly dependent on AI tools for learning and evaluation. [9] stressed the importance of educational policies that promote a balanced approach by integrating technology while nurturing essential cognitive abilities in students.

Table 3 Strategic Goals, Challenges, and Ethical Considerations of AI in Higher Education

<i>Identify the Strategic Goals and Benefits of Integrating AI</i>	<i>Investigate the Challenges and Barriers Faced by Higher Education Institutions in Adopting and Implementing AI</i>	<i>Examine the Ethical Considerations Associated with the Use of AI in Higher Education</i>
1. Enhanced Learning Experiences: AI chatbots improve student engagement and learning outcomes by providing immediate feedback	1. Technological Resource Constraints: Numerous educational institutions struggle with limited technological capabilities, which impedes the successful adoption of AI	1. Data Privacy Concerns: The gathering of student-related data poses substantial privacy risks, making it essential to implement strong safeguards for data protection
2. Operational Efficiency: Administrative tasks can be streamlined through AI, enabling teachers to dedicate more time to instruction	2. Resistance from Educators: Faculty members' doubts about AI technologies can hinder their willingness to implement them	2. Algorithmic Bias: AI systems may inadvertently sustain existing biases if not subjected to rigorous monitoring
3. Support for Sustainable Development: AI integration aligns with global educational goals and enhances access to quality education	3. Lack of Training Programs: The effective utilization of AI technologies is hampered by inadequate training for both educators and students	3. Over-reliance on Technology: Students' ability to think critically may be reduced by overreliance on AI technologies

4 CONCLUSIONS

Artificial intelligence integration within higher education institutions in sub-Saharan Africa offers considerable prospects for improving educational quality and operational effectiveness and poses significant challenges concerning infrastructure and ethical issues. Overcoming these obstacles is crucial for unlocking AI's transformative capabilities in the region's educational framework. A comprehensive strategy that includes training, infrastructure enhancement, and implementation of ethical guidelines is crucial for the successful incorporation of AI technology.

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REFERENCES

- [1] M. Hernandez-de-Menendez, C. Escobar Díaz, and R. Morales-Menendez, "Technologies for the future of learning: state of the art," *International Journal on Interactive Design and Manufacturing*, vol. 14, no. 2, pp. 683-694, 2020, doi: 10.1007/s12008-019-00640-0.
- [2] Abulibdeh, E. Zaidan, and R. Abulibdeh, "Navigating the confluence of artificial intelligence and education for sustainable development," *Journal of Cleaner Production*, vol. 437, art. no. 140527, 2024, doi: 10.1016/j.jclepro.2023.140527.
- [3] M.L Owoc, A. Sawicka and P.Weichbroth, "Artificial Intelligence Technologies in Education: Benefits, Challenges and Strategies of Implementation," *IFIP Advances in Information and Communication Technology*, vol. 599, pp .37-45 ,2021. doi: 10.1007/978-3-030-85001-2_4
- [4] J. E. McLaughlin, P. J. White, J. Khanova, and E. Yuriev, "Flipped Classroom Implementation: A Case Report of Two Higher Education Institutions in the U.S.", *Computers in the Schools*, vol. 33, no. 1, pp. 24-35, 2016, doi: 10.1080/07380569.2016.1137734.
- [5] P. Rajabi, P. Taghipour, D. Cukierman and T. Doleck, "Exploring ChatGPT's impact on post-secondary education: A qualitative study," *ACM International Conference Proceeding Series*, vol .9, pp 1 -6 ,2023, doi: 10.1145/3593342.3593360.
- [6] C. Huang and K. Yu, "Research on the innovation of college music teaching mode based on artificial intelligence," *Journal of Physics: Conference Series*, vol .1915, no.2, art.no .22051, 2021, doi: 10.1088/1742-6596/1915/2/022051.10
- [7] N. Sandu and E. Gide, "Adoption of AI-chatbots to enhance student learning experience in higher education in India," in *Proceedings of the 18th International Conference on Information Technology Based Higher Education and Training (ITHET)*, Magdeburg, Germany, 2019, pp. 1-6, doi: 10.1109/ITHET46829.2019.8937382.
- [8] S. Kozak, E. Ruzicky, J. Stefanovic, F. Schindler, "Research and education for industry 4 .0: Present development," in *Proceedings of the 29th International Conference on Cybernetics and Informatics*, Lazy pod Makytou, Slovakia, Jan 31 - Feb 3, 2018, pp .1 -6, doi: 10.1109/CYBERI.2018.8337556
- [9] R. A. Howell, "Engaging students in education for sustainable development: The benefits of active learning," *Journal of Cleaner Production*, vol. 325, art. no. 129318, 2021, doi: 10.1016/j.jclepro.2021.129318.